Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

fancyhdr xcolor array hyperref graphicx,amssymb,amsmath,multicol

[L]Short title, Semester [LE, RO]1

Header should include a short title and semester.
Do NOT include your name.

Template for NASA Keck Science Program

[the first 2 pages of this template contain general information and instructions; please remove them before submitting your proposal]

The NASA Keck proposal review is now using a Dual Anonymous Proposal Review (DAPR) process. We have prepared Guidelines for Proposers to assist with the preparation of your anonymous proposal. **Use of this template is required.**

In addition to this anonymously written Science Program, each proposal must also include a non-anonymized Expertise and Access (E&A) document prepared using this template available in Word or Latex. Further information on both the E&A document and the Science Program document can be found in the Call for Proposals.

Email us at keckcfp@ipac.caltech.edu with any questions.

**** SCIENCE PROGRAM PAGE LIMITS ****		
1. Science Case	up to 2 pages	
2. Instrument Request	up to 1 page	
3. Figures and Tables	up to 1 page	
4. References	up to 1 page	
Total for Science Plan	**up to 5 pages**	
5. Target list	no limit	
• The NExScI coversheet is generated via the submission website and includes fields for a title, PI and co-I names, abstract, NASA strategic relevance, and the Keck Observatory Archive data check. These should not be repeated in the proposal text. • The WMKO cover sheet is generated through the WMKO Observer Login Page. The WMKO proposal number should indicate the correct semester and start with an "N", indicating NASA as the allocating institution. You must upload a PDF of this coversheet when you submit your proposal. The NExScI and WMKO coversheets do not count against the above page limits.		

- This template is intended to outline the Science Program for a typical NASA Keck proposal. Section heading titles help make the proposal easier to read but are not required. Multiple sections can be put onto the same page, however shortening one section does not mean you can increase the page limit of another.
- Proposals must use 12-point font with 1-inch margins on all sides. Do not change the default font in this template since other fonts have different character spacing that can affect readability.
- Once your proposal is complete, use the online submission portal to submit the following PDF documents:
  - Anonymized Science Program
  - Non-anonymized Expertise and Access document
  - WMKO coversheet
  - Mission Support letters (if applicable)
- Following submission, you will receive a confirmation email with both the anonymized Science Program and the non-anonymized E&A document attached as PDFs. If you do not receive this confirmation email, please contact us at keckcfp@ipac.caltech.edu

1 Science Case (up to 2 pages)

Remember that all text in this template must conform to DAPR. Additionally, do not include an abstract or title here since these are captured in the coversheet upon proposal submission.

Specific points that should be addressed in the science case are detailed in the Call for Proposals. This section should demonstrate how the proposed science fits into NASA's strategic use of Keck time. Lastly, if you are requesting Mission Support status, this section should include a justification.

2 Instrument Request/Technical Justification (up to 1 page)

This section should contain details of your planned observations, descriptions of scheduling constraints, and data analysis plans. Specific points that should be addressed in this section are described in the Call for Proposals.

3 Figures, Tables (up to 1 page)

Figures and tables can be embedded within the narrative text or presented separately, but must not total more than one page of space. Captions may be in 10-point font.

4 References (up to 1 page)

References may be in 10 point font and should be formatted using the author's last name and year of cited publication, e.g. (Smith, 2015), (Smith and Smith, 2015) or (Smith et al., 2015).

Do not include references and links to papers in preparation or submitted that are stored on personal web pages. References to submitted papers on public archives (e.g., arXiv) are acceptable.

5 Target List (no page limit)

All proposals must include a complete, well-justified and, if appropriate, prioritized target list with sufficient information (magnitudes, coordinates) to determine scheduling within the semester, including which part of the night is requested if asking for less than full nights. *Proposals without such lists will be rejected.*

In the case of Targets of Opportunity or sources drawn from on-going surveys or very large samples of sources (e.g. from deep extragalactic fields), a description of the potential targets must be provided in sufficient detail for the Telescope Allocation Committee (TAC) to make a thorough review.